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Shared Task
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PSGS@2026: From Figures to Citation Graphs: QLoRA Based Multimodal Linking for AstroCLIMB

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TL;DR SUMMARY

- **Task:** Classify the relation between two astronomy figures or captions (`same_figure`, `same_paper`, `related_papers`, `unrelated_papers`) without DOI or metadata.
- **Methodology:** QLoRA finetuning of `google/gemma-4-E4B-it` on 8,200 competition training rows.
- **Key Result:** Validation macro F1 improves from **0.4408** to **0.5347**. The system achieves **0.64282 F1** on all 10,000 official test pairs.

1. INTRODUCTION

AstroCLIMB (WASP 2026 shared task) asks systems to predict one of four relations between a pair of figures and captions: `same_figure`, `same_paper`, `related_papers`, or `unrelated_papers`.

Models must rely on visual–textual content alone. Zero-shot vision-language models perform moderately well on `same_figure` but fail on the citation classes (`related_papers` near zero), which motivates for QLoRA finetuning.

2. METHODOLOGY

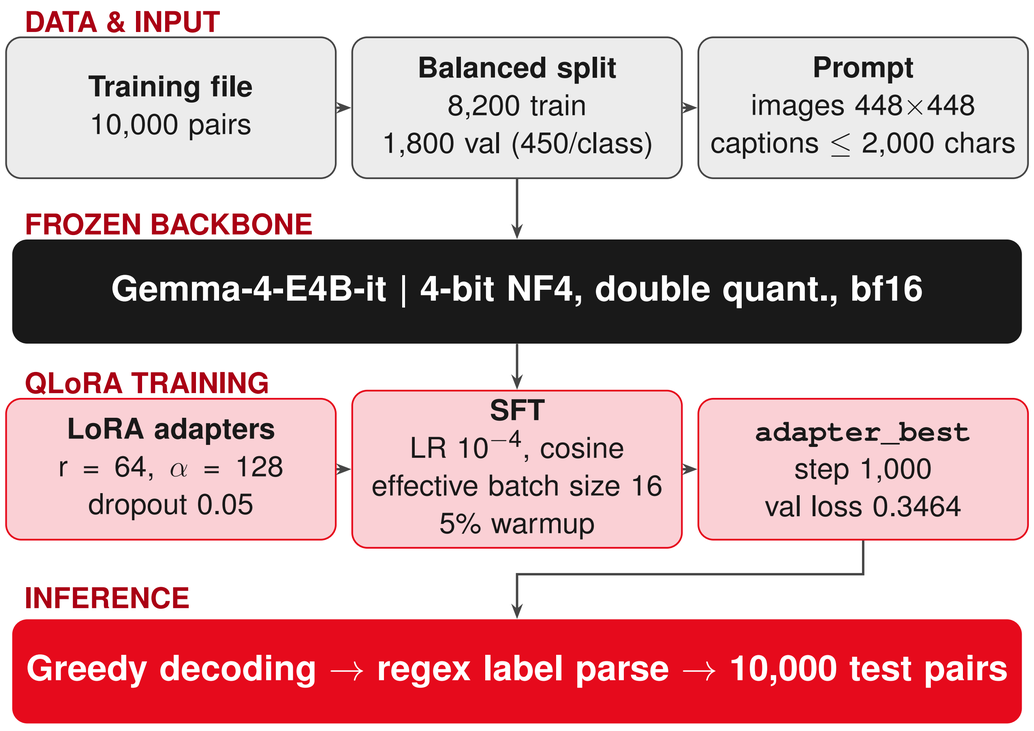


Figure 1: Finetuning and inference pipeline of the submitted system.

3. ERROR ANALYSIS

- `same_figure`: precision 0.9945 but recall 0.4000. 256 of 270 misses are labeled `same_paper`.
- Figure–figure `related_papers` pairs are the hardest subset (validation recall 0.0867).
- The test set shows the same pattern with a `same_figure` of precision 0.9386, recall of 0.6420 and `related_papers` F1 of 0.4535.

4. EXPERIMENTAL RESULTS

Metric: Macro F1 and per-class F1 on the balanced 1,800-pair validation split. The submitted system is also scored on the official 10,000-pair test set.

Run (F1)	sf	sp	rp	up
Gemma-4-E4B ZS	0.7760	0.4108	0.0172	0.5593
Gemma-4-E4B FS	0.7228	0.2047	0.0505	0.5918
SmolVLM-500M ZS	0.0128	0.0088	0.3333	0.3341
Gemma-4-E4B FT	**0.5705**	**0.5311**	**0.3508**	**0.6864**

sf = `same_figure`, sp = `same_paper`, rp = `related_papers`, up = `unrelated_papers`; ZS = zero-shot, FS = few-shot, FT = finetuned.

Model	Mode	Macro F1	Acc.
Gemma-4-E4B	Zero-shot	0.4408	0.5206
Gemma-4-E4B	Few-shot + thinking	0.3925	0.5050
SmolVLM-500M	Zero-shot	0.1723	0.2511
Qwen3-VL-8B-Thinking	Zero-shot	0.1068	0.2461
Gemma-4-E4B + QLoRA	**Finetuned**	**0.5347**	**0.5483**

Finetuned model	Val. loss	Official test F1
Gemma-4-E4B, LoRA-64	**0.3464**	**0.64282**
SmolVLM-500M, LoRA-32	0.3029	0.23675

5. CONCLUSION

QLoRA finetuning of Gemma-4-E4B-it improves validation macro F1 from 0.4408 to 0.5347 and `related_papers` F1 from 0.0172 to 0.3508, with an official test set F1 score of **0.64282**, at the cost of `same_figure` F1 (0.7760 to 0.5705). Few shot thinking (0.3925) does not beat zero-shot.

Limitations. No caption-only or image-only ablations. Validation is in-distribution (split by pair row).

Future work. Future works includes exploring pair-type-aware objectives for `related_papers` and finetuning thinking-capable backbones.

Released. `huggingface.co/ShivRamSaud/gemma4-e4b-qlora astroclimb-v2-fresh`